BFS Python Code

Formatted copy of the uploaded bfs.py file

```
def bfs(graph, start, goal):
    queue = [start]
    visited = [start]
    parent = {start: None}

    while queue:
        node = queue.pop(0)
        print("Visiting:", node)

        if node == goal:
            path = []
            while node is not None:
                path.append(node)
                node = parent[node]
            path.reverse()
            return path

        for child in graph[node]:
            if child not in visited:
                visited.append(child)
                parent[child] = node
                queue.append(child)

    return None


graph = {
    'A': ['B', 'C'],
    'B': ['A', 'D', 'E'],
    'C': ['A', 'F', 'G'],
    'D': ['B'],
    'E': ['B', 'H'],
    'F': ['C'],
    'G': ['C', 'I'],
    'H': ['E'],
    'I': ['G']
}

start = 'A'
goal = 'I'

path = bfs(graph, start, goal)

if path:
    print("Path:", " -> ".join(path))
    print("Path cost:", len(path) - 1)
else:
    print("No path exists")
```